MILITARY SPECIFICATION SHEET

ELECTRON TUBE, POWER

TYPES 6283 AND 8500

The complete requirements for procuring the electron tube described herein shall consist of this document and the issue in effect of Specification MIL-E-1.

DESCRIPTION: Tetrode
See figures 1 and 2
Mounting position: Any
Weight: 1 pound nominal

ABSOLUTE RATINGS: F1 = 400 MHz and F2 = 900 MHz See notes 1 and 2

Parameter: / Unit:		Ef V	Eb kVdc	Ec1 Vdc	Ec2 Vdc	Ic1 mAdc	Ib mAdc
Maximum:							
C Teleg:	(900 MHz)	6.8	1.6	-100	320	50	300
B RF:	(400 MHz)	6.8	2.0	----	320	50	250
TEST CONDITIONS:		6.3	1.0	Adjust	Adjust	----	----

ABSOLUTE RATINGS:

Parameter: / Unit:		Cooling (See note 1)	Pg1 W	Pg2 W	Pp W	Pi W	tk sec
Maximum:							
C Teleg:	(900 MHz)	---	2	15	480	480	60
B RF:	(400 MHz)	---	---	5	500	500	60
TEST CONDITIONS:		---	---	---	---	---	60

GENERAL

Qualification - Required

(E) Denotes Changes

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METHOD	TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS Min	LIMITS Max	UNIT
	Qualification							
(F) 2214	Power output (1)	Ef = 5.5 V; Eb = 2,000 Vdc; Ec2 = Adjust; It = 350 mAdc; Ic1 = [illegible] mAdc; Ec1 = [illegible] Vdc; F = 350-400 MHz; (See note 4)	---	---	Po	[illegible]	---	W
----	Life-test(1)provisions	Power output (1); t = 500 hours	---	---	---	---	---	--
----	Life-test(1)end points	Pulsing emission sinusoid	---	---	Is	40	---	a
		Power output (1)	---	---	Po	135	---	W Useful
	Quality conformance inspection, part 1							
1501	Heater current		0.65	II	If	3.5	4.0	A
1256	Total grid current	Eb/It = 300 mAdc; Eb = 2,000 Vdc; Ec2 = 300 Vdc; (See note 7)	0.65	II	Ic1	---	-7.0	µAdc
1201	Electrode voltage (grid) (cutoff)	Ec2 = 300 Vdc; Ec1/It = 10 mAdc	0.65	II	Ec1	-[illegible].0	-[illegible].0	Vdc
1231	Pulsing emission sinusoid	Ec1 = ec2 = eb = 150 V; tA = [illegible] µs (see note 5)	0.65	II	Is	60.0	---	a
(E) 2214	Power output (1)	Eb = 1,500 Vdc; Ec2 = Adjust It = 300 mAdc; Eg = 2,000 volts; Ic1 = 20[illegible] Adc; F = 350-400 MHz; (See note 4)	0.65	II	Po Ic2	140 ---	--- 15	W (Useful) mAdc
	Quality conformance inspection, part 2							
1021	Low-frequency vibration	No voltage	---	---	---	---	---	---
1201	Electrode voltage (1) (grid)	Ec2 = 300 Vdc; It = 300 mA	---	---	Ec1	-8	-15	Vdc
1201	Electrode voltage (2) (grid)	Ec2 = 250 Vdc; It = 200 mA	---	---	Ec1	-5	-11	Vdc
1201	Electrode voltage (3) (grid)	Ec2 = 300 Vdc; It = 250 mA	--	---	Ec1	-5	-12	Vdc
1316	Amplification factor	See note 5	---	---	Mu	11	17	
1306	Transconductance	See note 6	---	---	Sm	20,000	---	µmhos
1431	Direct-interelectrode capacitance	g1 to g2, {528 / 8500	---	---	Cin	17	19.5	pF
		anode grounded	---	---	Cin	18	21	pF
		g1 to g2, cathode grounded	---	---	Cout	6.1	6.7	pF
	Quality conformance inspection, part 2							
----	Life-test(2)provisions	Group D; Power output (1); Ef = 5.5 V; t = 500 hrs.	---	---	---	---	--	---
----	Life-test(2)end points	Power output (1)	---	---	Po	135	---	W Useful
					Ic2	---	20	mAdc
		Pulsing emission sinusoid (See note 5)	---	---	Is	40	---	a

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NOTES:

1. Airflow required for cooling at sea level through the radiators shall meet the following:

Anode dissipation	500 W
Airflow	17 CFM (min)
Static pressure	0.98 IN. (H_2O) (Approx)
Heater-cathode seals	1.0 CFM (min)
Screen-grid control-grid seals	1.0 CFM (min)
Incoming air temperature	45°C (max)
Radiator hot temperature (at fin adjacent to anode seal)	250°C (max)
Tube temperature (at any point)	200°C (max)

Cooling shall be applied prior to and during application of any voltages and shall be maintained for 1 minute after the removal of all voltages. Provision shall be made for unobstructed passage of cooling air between radiator fins and between the anode terminal and adjacent radiator fin. The volume of cooling air indicated for various seals is only approximate. Distribution of cooling air may vary with the cavity configuration about the tube. For most satisfactory operation, the maximum temperature at any point on the tube shall be below 200°C.

2. The cathode of the tube, because of transit time effects which raise the temperature of the cathode is subject to considerable back bombardment in ultra-high frequency service. The amount of heating due to bombardment is a function of the operating conditions and frequency, and must be compensated for by a reduction of the heater input to prevent over heating of the cathode with resulting short life. For long life, tube type 8283 and 8500 should be put in operation with rated heater voltage. After the circuit has been adjusted for proper tube operation the heater voltage should be reduced to a value slightly above that at which circuit performance is affected. In any case, it is important from a tube life standpoint to keep the filament voltage at as low a level as possible consistent with required performance. However, the filament voltage should not be reduced below 5.5 volts. At a frequency of 900 MHz and with typical operating conditions the heater voltage can be reduced to approximately 5.5 volts. At lower frequencies, the reduction will be less. Minor circuit readjustment may be necessary after this adjustment. The procedure for determining proper heater power should be repeated periodically.

3. Applied voltage shall be approximately sinusoidal. The pulse duration 2 to 3.5 µsec, prr = 60.

4. Any frequency within the specified range may be used.

5. This test shall be for screen grid to control-grid amplication factor and shall be calculated by dividing the difference in Ec2 (50V) by the difference in Ec1 as read from the electrode voltage (1) (grid) and electrode voltage (2) (grid) tests.

6. This test shall be calculated by dividing 50,000 by the difference in Ec1 as read from the electrode voltage (1) and electrode voltage (3) (grid) tests.

7. This test to be the first test performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 80

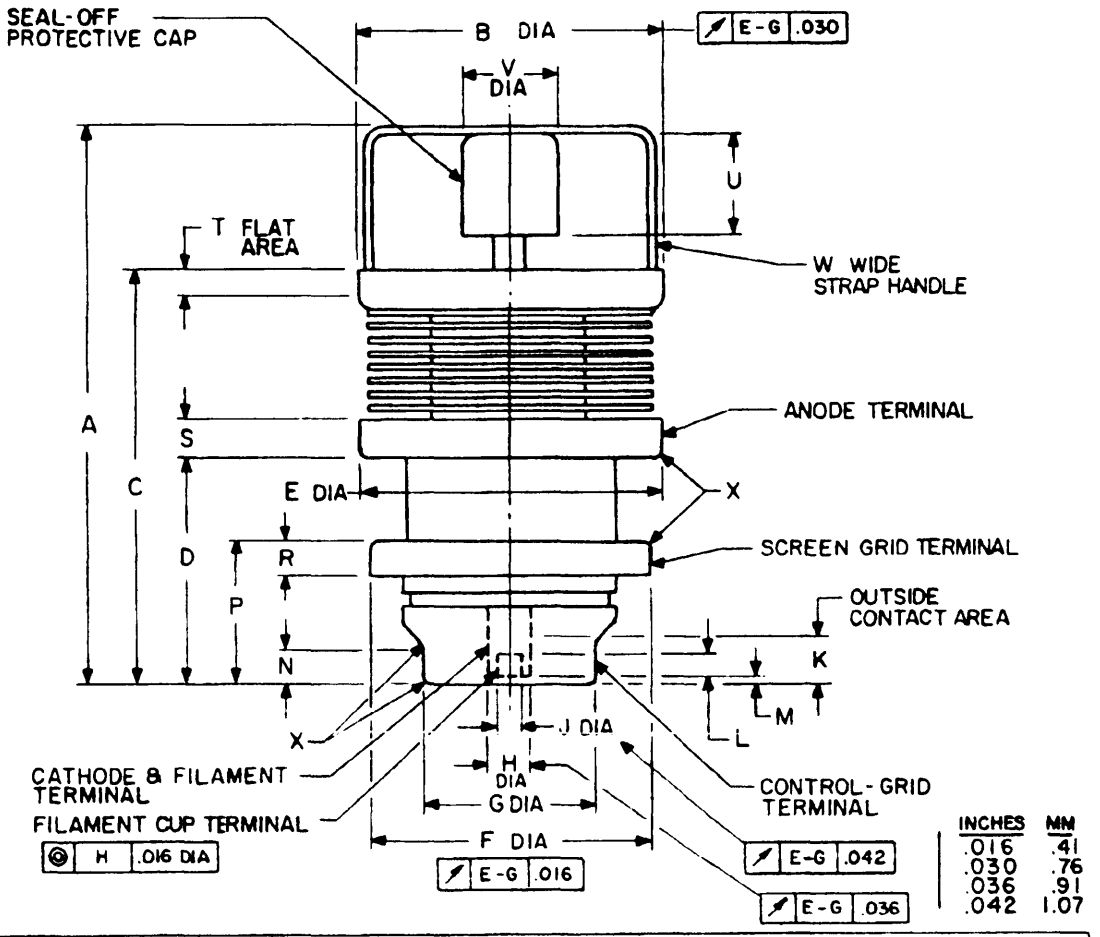
Review activities:
Army - EL
Air Force - 11, 17

User activities:
Army - WC
Navy - MC, CG, AS, DS, SH
Air Force - None

Preparing activity:
Air Force - 80

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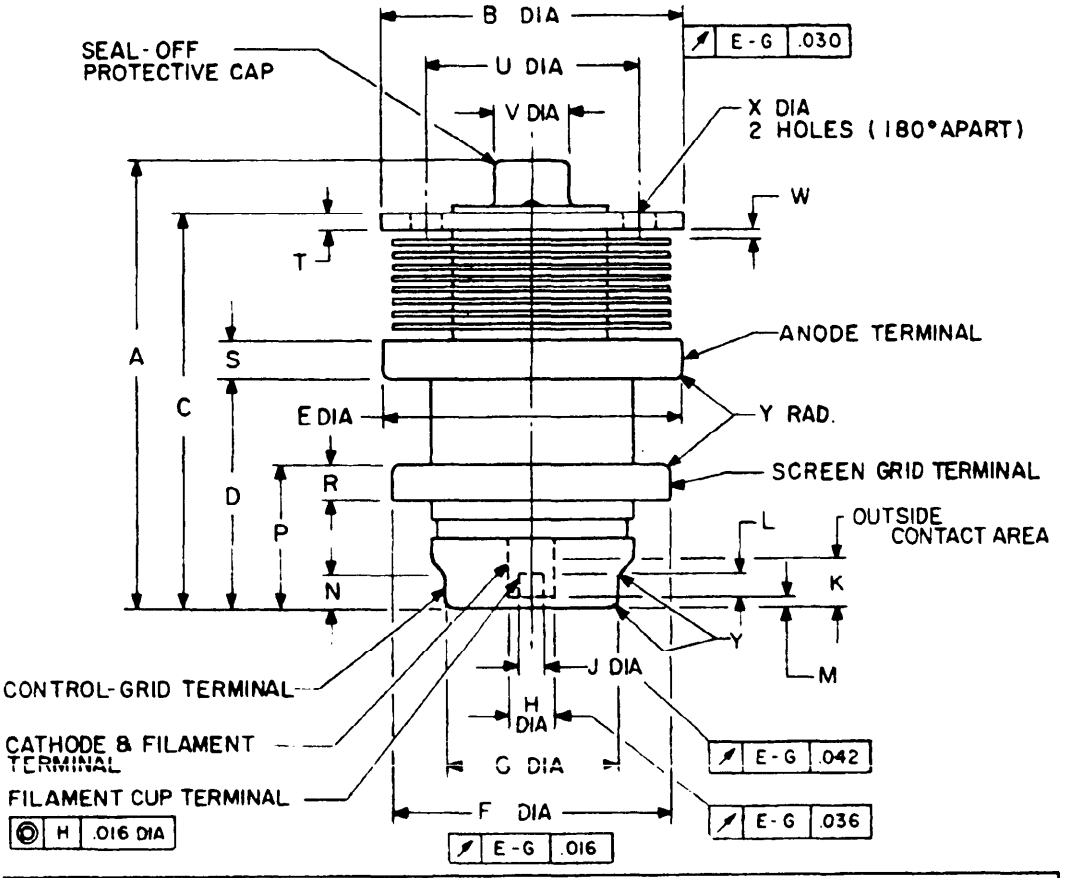


DIMENSIONS													
LTR	INCHES		MILLIMETERS		LTR	INCHES		MILLIMETERS		LTR	INCHES		MILLIMETERS	
	MIN	MAX	MIN	MAX		MIN	MAX	MIN	MAX		MIN	MAX	MIN	MAX
QUALITY CONFORMANCE INSP., PART 2					H	.320	.328	8.13	8.33	S	.281	.312	7.14	7.92
A	4.094	4.344	103.99	110.34	J	.181	.191	4.60	4.85	T	.188		4.78	
B	2.303	2.323	58.50	59.00	K	.375		9.53		U	.688	.813	17.48	20.65
C	3.047	3.234	77.39	82.14	L	.188		4.78		V	.625	.750	15.88	19.05
D	1.672	1.765	42.47	44.83	M	.063	.125	1.60	3.18	W	.344	.406	8.74	10.31
E	2.271	2.291	57.68	58.19	N	.250		6.35		X		.047		1.19
F	2.115	2.135	53.72	54.23	P	1.063	1.125	27.00	28.58					
G	1.302	1.322	33.07	33.58	R	.250	.313	6.35	7.95					

NOTES:
1) METRIC EQUIVALENTS (TO THE NEAREST .01 MM) ARE GIVEN FOR GENERAL INFORMATION ONLY AND ARE BASED UPON 1 INCH = 25.4 MM.
2) BOTTOM OF CUP MUST NOT BE USED AS A SOCKET STOP.

FIGURE 1. Outline drawing of electron tube type 6283

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DIMENSIONS									
LTR	INCHES		MM		LTR	INCHES		MM	
	MIN	MAX	MIN	MAX		MIN	MAX	MIN	MAX
QUALITY CONFORMANCE INSPECTION, PART 2					M	.063	.125	1.60	3.18
A	3.297	3.453	83.74	87.71	N	.250		6.35	
B	2.303	2.323	58.50	59.00	P	1.063	1.125	27.00	28.58
C	2.921	3.047	74.19	77.39	R	.250	.313	6.35	7.95
D	1.672	1.765	42.47	44.83	S	.281	.312	7.14	7.92
E	2.271	2.291	57.68	58.19	T	.109	.141	2.77	3.58
F	2.115	2.135	53.72	54.23	U	1.615	1.635	41.02	41.53
G	1.302	1.322	33.07	33.58	V	.500	.625	12.70	15.88
H	.320	.328	8.13	8.33	W	.063		1.60	
J	.181	.191	4.60	4.85	X	.256	.276	6.50	7.01
K	.375		9.53		Y		.047		1.19
L	.188		4.78						

NOTES:
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2) BOTTOM OF CUP MUST NOT BE USED AS A SOCKET STOP.

FIGURE 2. Outline drawing of electron tube type 8500

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

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